

SOLUTION REQUIREMENTS

Date	26 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

The AI-Powered Flood Prediction and Alert System shall securely collect real-time and historical weather data to predict flood risk using Machine Learning. It shall classify flood risk levels, display results on an interactive dashboard, and send timely alerts through SMS, email, and in-app notifications. The system shall provide fast, accurate, and reliable predictions while ensuring data security, high availability, and scalability. It shall also offer a responsive, user-friendly interface that is easy to maintain and compatible with modern devices and web browsers.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	User Registration & Login	Users should be able to register, log in securely, and manage their profiles.	High

2	AI-Based Flood Prediction	Use a Machine Learning model to predict flood risk based on rainfall, river level, humidity, and historical data.	High
3	Alert Notifications	Send instant alerts through SMS, Email, and in-app notifications when flood risk is detected.	High
4	Risk Classification	Classify flood risk into Low, Medium, and High based on predefined thresholds and ML predictions.	Medium

Step 2: Non-Functional Requirements (NFR)

1	Performance & Speed	System should generate flood predictions and alerts quickly with minimal delay.	Prediction generated within 5 seconds and page loads within 2 seconds .
2	Scalability	System should support increasing users and large weather datasets efficiently.	Supports 10,000+ concurrent users .

3	Security & Data Privacy	Secure user information and weather data using encryption and authentication.	End-to-end encryption and secure login with HTTPS.
4	Reliability & Availability	System should remain available during emergency situations with minimal downtime.	99.9% uptime.
5	Usability & Accessibility	User-friendly interface accessible on desktop and mobile devices.	Responsive design with easy navigation.
6	Maintainability	Code should be modular, well-documented, and easy to update.	Proper documentation and clean code architecture.